WEEK 7 · FRIDAY · PRACTICAL LAB



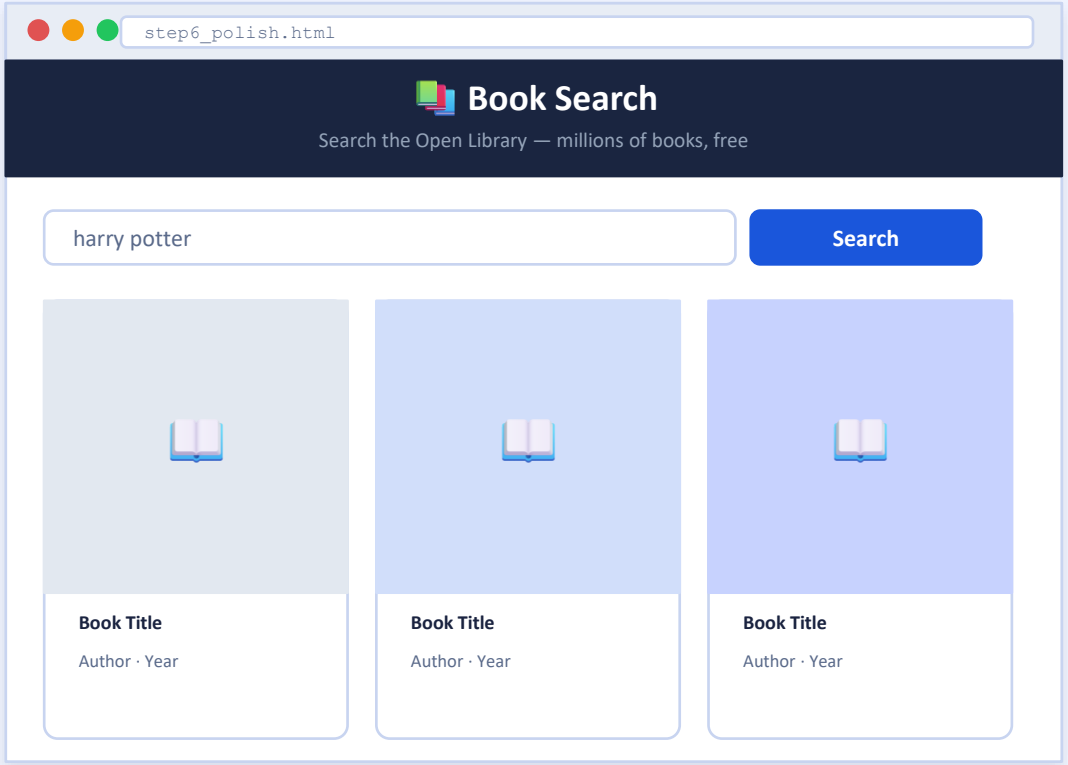
Book Search App

Apply everything from Weeks 1–6 in one real project

Semantic HTML | CSS Variables | Flexbox / Grid | JS DOM | Fetch API | async / await

WHAT YOU WILL BUILD TODAY

A live book search app



User types a book title

↓ fetch() sends request to Open Library API

↓ JSON response arrives (title, author, cover…)

↓ JS creates a card element for each book

↓ Cards appear in the grid

Semantic HTML + CSS

addEventListener

fetch + async/await

createElement

CSS Grid

THE API YOU WILL USE

Open Library Search API

Free · No API key · No account needed · Works in browser

```
https://openlibrary.org/search.json?q=harry+potter&limit=12
```

Response shape (data.docs[0])

```
.title string
→  "Harry Potter and the Philosopher's Stone"
```

```
.author_name[0] string
→  "J. K. Rowling"
```

```
.first_publish_year number
→  1997
```

```
.cover_i number?
→  8091016    (may be undefined — handle it!)
```

Build the cover URL from the ID

```
https://covers.openlibrary.org/b/id/ cover_i -M.jpg
```

Size options: `-S.jpg` (small) `-M.jpg` (medium ✓ use this) `-L.jpg` (large)

⚠ **cover_i may be undefined — always check before building the URL!**

If it's missing, show a placeholder emoji 📚 instead of a broken image.

```
const imgHtml = book.cover_i
  ? '<img src="https://covers.openlibrary.org/b/id/'
      + book.cover_i + '-M.jpg">'
  : '<div class="no-cover">📚</div>';
```

TODAY'S LAB STRUCTURE

6 steps → one complete app

1

HTML Structure

Write semantic HTML — header, form, results section

Semantic HTML

2

CSS Layout & Cards

CSS variables, Flexbox search bar, Grid card layout

CSS

3

JS DOM — Wire the Form

getElementById, addEventListener, preventDefault

JS DOM

4

Fetch — Call the API

fetch() → check res.ok → parse JSON

Fetch API

5

Render — Display Cards

createElement → set content → appendChild

JS DOM

6

Bonus — Loading & Errors

Skeleton cards, disabled button, empty state message

try/finally UX

BEFORE YOU BEGIN

Lab ground rules

Solution file is the last resort

Try each step yourself first. The solution is there if you are truly stuck — not as a shortcut. You learn nothing by reading code you didn't write.

No copy-paste from GPT / ChatGPT

Typing code yourself builds the muscle memory. If you don't understand a line, you can't debug it later — and the midterm has no internet.

Ask for help early

Raise your hand when you are stuck for more than 3 minutes. The instructor is here to unstick you — don't lose 10 minutes on a typo.

Test each step in the browser

Open DevTools (F12) and check the Console after every step. Real feedback from the browser is faster than re-reading the code.

Open this file to start:

week7_lab / index.html

① Open

Double-click index.html
in the week7_lab folder

② Read

Click Step 1 → read the
task comment at the top

③ Code

Write your code, save,
refresh browser to test

You have 80 minutes. Aim for Step 5 as the minimum — Step 6 is a bonus challenge.

Use the API Explorer (api_explorer.html) to understand the JSON data shape before coding

DevTools (F12) → Console tab → check for errors after every step